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$$X = \hat{\alpha}_1 + \hat{\beta}_1 Y \quad \hat{\beta}_1 = \frac{\hat{\sigma}(X, Y)}{\text{Var}(Y)} \quad Y = \hat{\alpha}_2 + \hat{\beta}_2 X$$

$$\Leftrightarrow Y = -\frac{\hat{\alpha}_1}{\hat{\beta}_1} + \frac{1}{\hat{\beta}_1} X \quad \hat{\beta}_2 = \frac{\hat{\sigma}(Y, X)}{\text{Var}(X)}$$

$$\hat{\beta}_1 \hat{\beta}_2 = \hat{\rho}_{XY}^2 < 1$$

If $\hat{\beta}_2 = 2.5$ and $\frac{1}{\hat{\beta}_1} = 1.6$ then $\hat{\beta}_1 \hat{\beta}_2 = \frac{2.5}{1.6} > 1$ ✗

If $\hat{\beta}_2 = 1.6$ and $\frac{1}{\hat{\beta}_1} = 2.5$ then $\hat{\beta}_1 \hat{\beta}_2 = \frac{1.6}{2.5} < 1$ ✓

These are the only possibilities.

So $Y = .4 + 1.6X$ is the regression line of Y on X

and $Y = -.5 + 2.5X$ - - - X on Y .

⑥ We have to use the regression line of Y on X for this.

$$\hat{E}(Y|X) = .4 + 1.6X \quad \text{For } X=20, \text{ this is } .4 + 1.6 \times 20 = 32$$

⑦ Since $\hat{\beta}_1$ and $\hat{\beta}_2$ are positive, $\hat{\rho} = \sqrt{\hat{\beta}_1 \hat{\beta}_2} = \sqrt{\frac{1.6}{2.5}} = .8$

$$\left. \begin{array}{l} \bar{Y} - 1.6\bar{X} = .4 \\ \bar{Y} - 2.5\bar{X} = -.5 \end{array} \right\} \text{Both regression lines have to pass through } (\bar{X}, \bar{Y}).$$

Solving these we get $\bar{X} = 1, \bar{Y} = 2$.

$$\frac{\hat{\beta}_1}{\hat{\beta}_2} = \frac{\text{Var}(X)}{\text{Var}(Y)} \quad \text{Ratio of standard deviations is}$$

$$\sqrt{\frac{\hat{\beta}_1}{\hat{\beta}_2}} = \frac{1}{\sqrt{2.5 \times 1.6}} = \frac{1}{2}$$

[2 is also acceptable. as it is $\frac{\text{SD}(Y)}{\text{SD}(X)}$]

2a) Multiple Regression

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + E$$

Y : amount of credit availed

X_1 : gender

X_2 : educational level

X_3 : caste.

$$E(E) = 0, \text{var}(E) = \sigma^2$$

$$E \perp (X_1, X_2, X_3).$$

X_i are categorical and can be coded with indicator variables.

b) ANOVA one way with different levels for the different castes and the response is ^{amount of} credit availed.

$$X_{ij} = \mu + \beta_i + E_{ij} \text{ for caste } i \text{ and individual } j \text{ in that case.}$$

$j = 1, \dots, n_i$

c) Logistic regression.

$$p_i = P(Y=1 | X_1, X_2, X_3)$$

$$\ln \frac{p_i}{1-p_i} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

Y, X_1, X_2, X_3 are as in part a).

~~② χ^2 test for $H_0: p_1 = p_2 = \dots = p_k$ where k is the number of castes.
(of goodness of fit.)
If n is the total number of observations, then
the expected count of i th cell is n/k for $i=1, \dots, k$.~~

③ χ^2 test of independence. Two way table. One way is loan pay back/not pay back. Other way is caste.

$$3. \textcircled{a} \hat{\beta}_1 = \frac{\sum (x_{1i} - \bar{x}_1) y_i}{\sum (x_{1i} - \bar{x}_1)^2}$$

$$\begin{aligned} E \hat{\beta}_1 &= \frac{\sum (x_{1i} - \bar{x}_1) (\alpha + \beta_1 x_{1i} + \beta_2 x_{2i})}{\sum (x_{1i} - \bar{x}_1)^2} \\ &= \frac{\alpha \sum (x_{1i} - \bar{x}_1) + \beta_1 \sum (x_{1i} - \bar{x}_1) x_{1i} + \beta_2 \sum (x_{1i} - \bar{x}_1) x_{2i}}{\sum (x_{1i} - \bar{x}_1)^2} \\ &= \frac{0 + \beta_1 \sum (x_{1i} - \bar{x}_1)^2 + \beta_2 \sum (x_{1i} - \bar{x}_1) x_{2i}}{\sum (x_{1i} - \bar{x}_1)^2} \\ &= \beta_1 + \beta_2 \frac{\text{Cov}(x_1, x_2)}{\text{Var}(x_1)} \end{aligned}$$

$$\text{Bias} = E \hat{\beta}_1 - \beta_1 = \beta_2 \cdot \frac{\text{Cov}(x_1, x_2)}{\text{Var}(x_1)}$$

~~① If x_1 and x_2 are uncorrelated (We do not need $\wedge$ (estimates) as x 's are fixed]~~

② The estimator is unbiased if the centered vectors of x_1 and x_2 are orthogonal.

[It is acceptable if students do not realize x 's are fixed and say x_1 is uncorrelated with x_2 or x_1 is indep of x_2].

4. Done in homework.

5. Done in homework.

	Df	SS	MS	F	P(>F)
Species	2	11.35	5.675	49.35	<2e-16
Residuals	147	16.905	0.115		

- (b) Y = Sepal width
 i denotes species. $i=1, 2, 3$.
 j denotes the identity of the flower sample within each species.
 $j=1, \dots, 50$.

Model: $Y_{ij} = \alpha + \beta_i + \epsilon_{ij}$

$H_0: \beta_1 = \beta_2 = \beta_3$

H_a : ~~One or~~ At least one of the β_i 's is different from the others.

Assumptions: ϵ_{ij} are indep. $E\epsilon_{ij}=0$, $\text{Var}(\epsilon_{ij})=\sigma^2$.
 ϵ_{ij} follow normal distn.

Test statistic = 49.35

Distr of test statistic is $F_{2,147}$.

p-value is very small.

We reject H_0 at $\alpha=5\%$ or $\alpha=1\%$.

Conclusion: There is sufficient evidence in the data to conclude that all three species do not have the same average sepal width.

Student Name _____

7) For $n=592$ individuals, we note hair color and eye color.

There are $i=1, 2, 3, 4$ values of hair color
and $j=1, 2, 3, 4$ - - - eye color.

The individual is counted in the (i, j) th cell if they have i th hair color & j th eye color.

The numbers n_{ij} in cell (i, j) denote the number of individuals with i th hair color & j th eye color.

p_{ij} is the probability of a ^{random} individual belonging to (i, j) th cell.

$p_{i.}$ = prob of a random individual having i th hair color.

$p_{.j}$ = - - - - - j th eye color.

$$H_0: p_{ij} = p_{i.} p_{.j} \quad i=1, \dots, 4, j=1, \dots, 4.$$

H_a : H_0 is not true.

Assumption: Individuals are selected independently of each other and with equal probability from the popn.

$$\text{Test statistic} = \sum_{i=1}^4 \sum_{j=1}^4 \frac{(n_{ij} - n_{i.} n_{.j} / n)^2}{n_{i.} n_{.j} / n}, \quad n_{i.} = \sum_j n_{ij}, \quad n_{.j} = \sum_i n_{ij}$$

$n=592$.

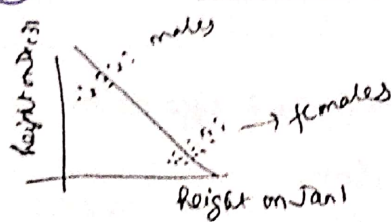
Distro of test statistic is χ^2 with 9 df.

Value of test statistic is 138.29

p -value is very small. So we reject H_0 .

Conclusion: Hair color and eye color are not independent attributes.

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Multiple regression of y (height on Dec 31) on x_1 (gender) and x_2 (height on Jan 1) will address this issue. In the plot we have done regression of y on x_2 only to get negative slope. ~~But~~

If we put $x_1 = \begin{cases} 1 & \text{male} \\ 0 & \text{female} \end{cases}$ then in the multiple regression

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

We shall get $\beta_1 > 0$ and $\beta_2 < 0$ for a fixed gender the height is increasing
 $\downarrow$ males taller than females

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1) $(y_1, \dots, y_6) \sim \text{Multinomial}_6(100, (\frac{1}{6}, \dots, \frac{1}{6}))$.

2) ~~$(y_1, y_2, y_3) \sim \text{Multinomial}_3(100, (\frac{1}{6}, \frac{1}{6}, \frac{1}{6}))$~~
 $(y_1, y_2, y_3, 100 - y_1 - y_2 - y_3) \sim \text{Multinomial}_4(100, (\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{2}))$

$$P(y_1 = y_1, y_2 = y_2, y_3 = y_3) = \frac{100!}{y_1! y_2! y_3! (100 - y_1 - y_2 - y_3)!} \left(\frac{1}{6}\right)^{y_1 + y_2 + y_3} \left(\frac{1}{2}\right)^{100 - y_1 - y_2 - y_3}$$

$(y_1, y_2, y_3, 80 - y_1 - y_2 - y_3) \sim \text{Mult}_4(80, (\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{2}{5}))$

$$P(y_1 = y_1, y_2 = y_2, y_3 = y_3) = \frac{80!}{y_1! y_2! y_3! (80 - y_1 - y_2 - y_3)!} \left(\frac{1}{5}\right)^{y_1 + y_2 + y_3} \left(\frac{2}{5}\right)^{80 - y_1 - y_2 - y_3}$$